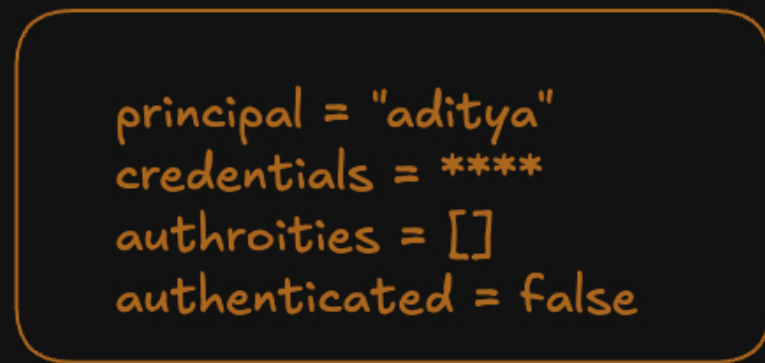
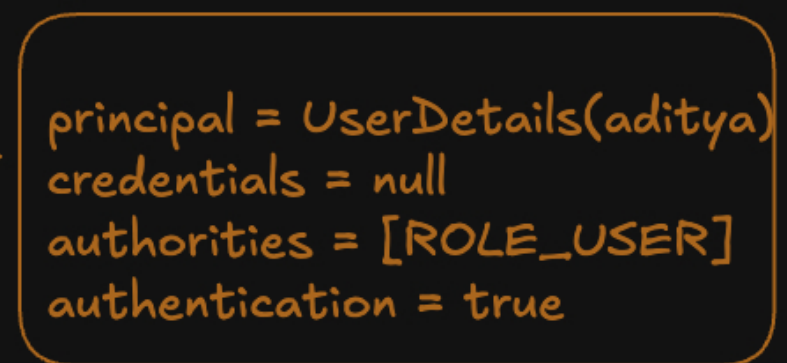


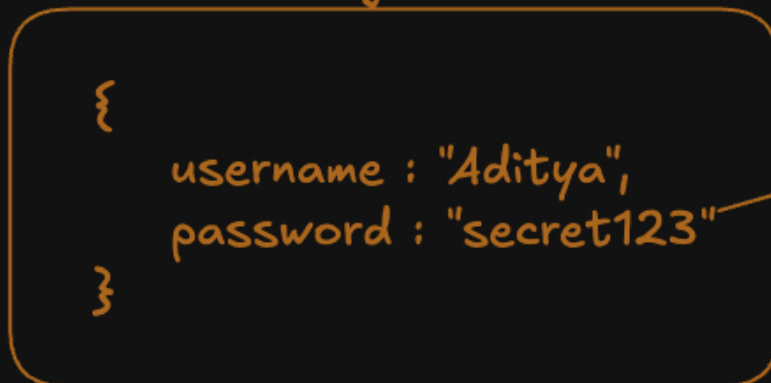


Authentication



Authentication

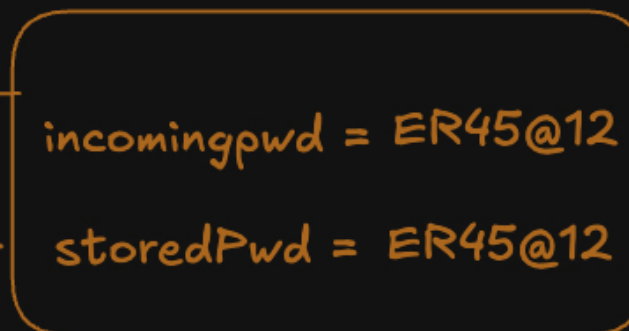
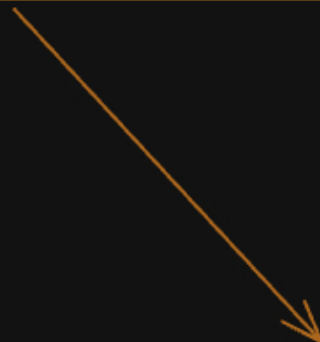
Login



ER45@12

UserDetails

UserDetailsService



Application

aditya : ER45@12

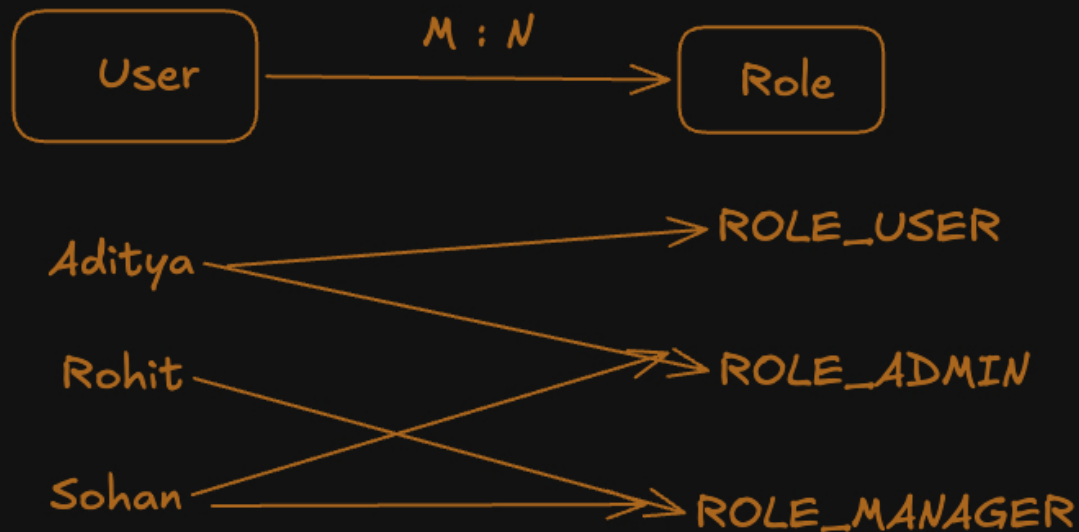


UserDetail

```
public interface UserDetails {  
  
    String getUsername();  
  
    String getPassword();  
  
    Collection<? extends GrantedAuthority> getAuthorities();  
  
    boolean isAccountNonExpired();  
  
    boolean isAccountNonLocked();  
  
    boolean isCredentialsNonExpired();  
  
    boolean isEnabled();  
}
```

User

```
class User {  
    String username;  
    String password;  
}
```



User

id	name
1	Aditya
1	Aditya

Role

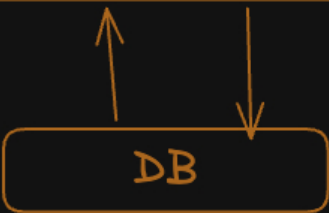
id	name
101	ADMIN
102	USER

User_role

user_id	role_id
1	101
1	102

Entities





User

```
POST /api/users
{
  username : "Aditya",
  password : "secret123"
}
```

username

password

username	pwd
Aditya	HER56/_+231

QWERTY

Raw	Bcrypt	x	y
password	QW123	ERT67	ERgh
aditya123	RE456		
12345	SDF56		
secret123	ER34@12		

Rainbow Table

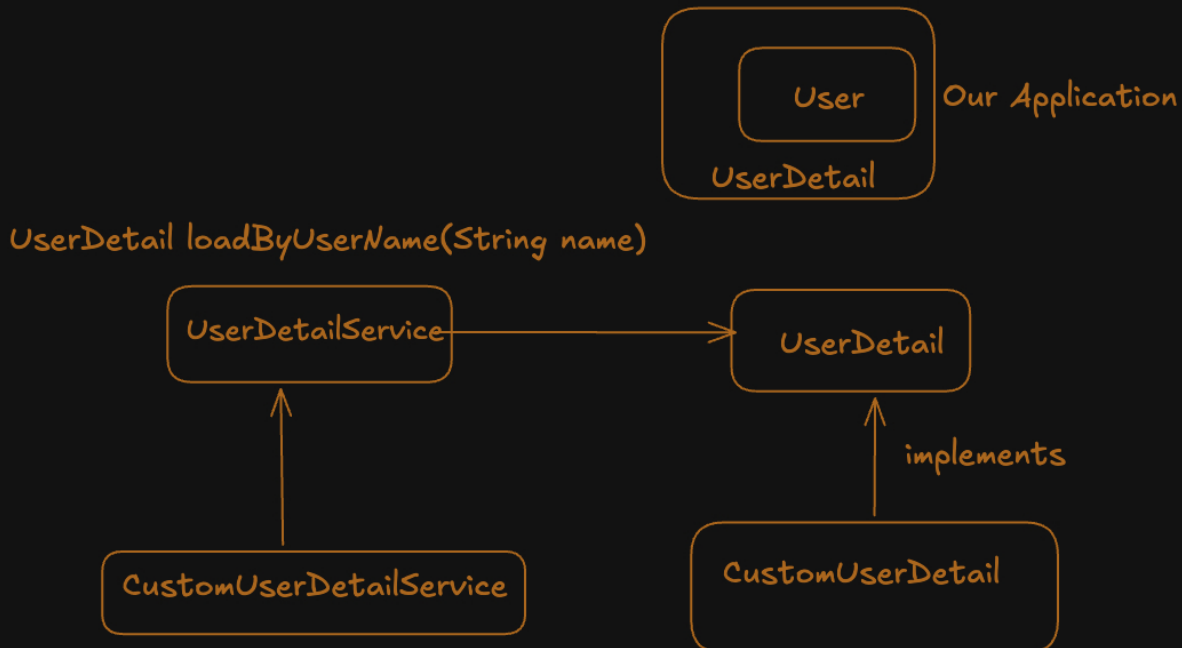


PasswordEncoder

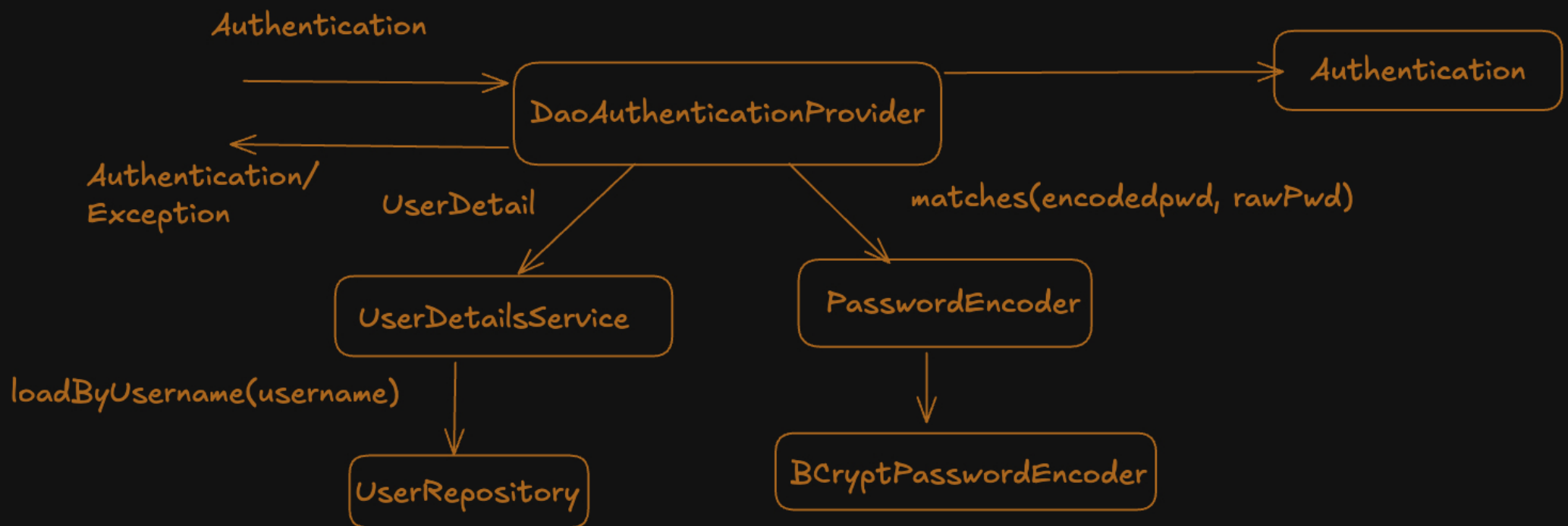
BCryptPasswordEncoder

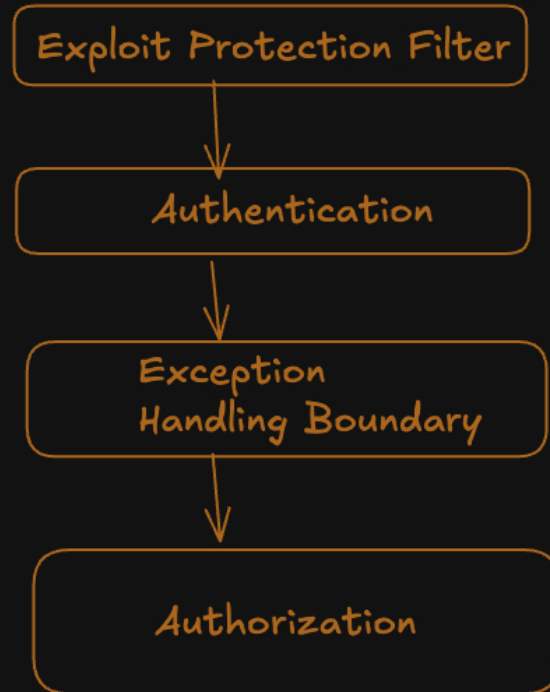
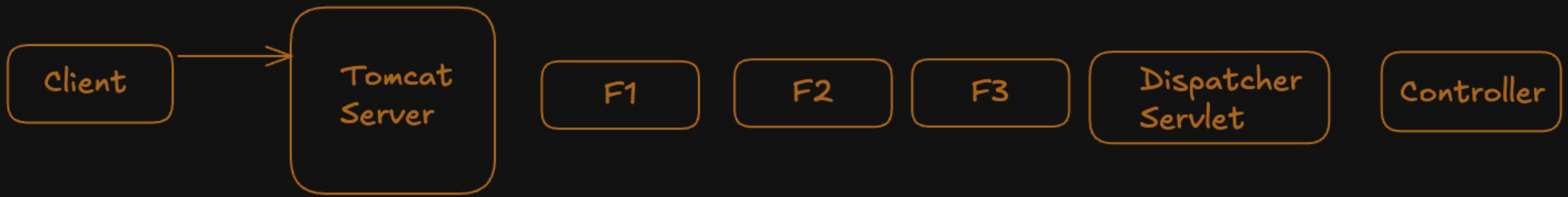
UserDetail

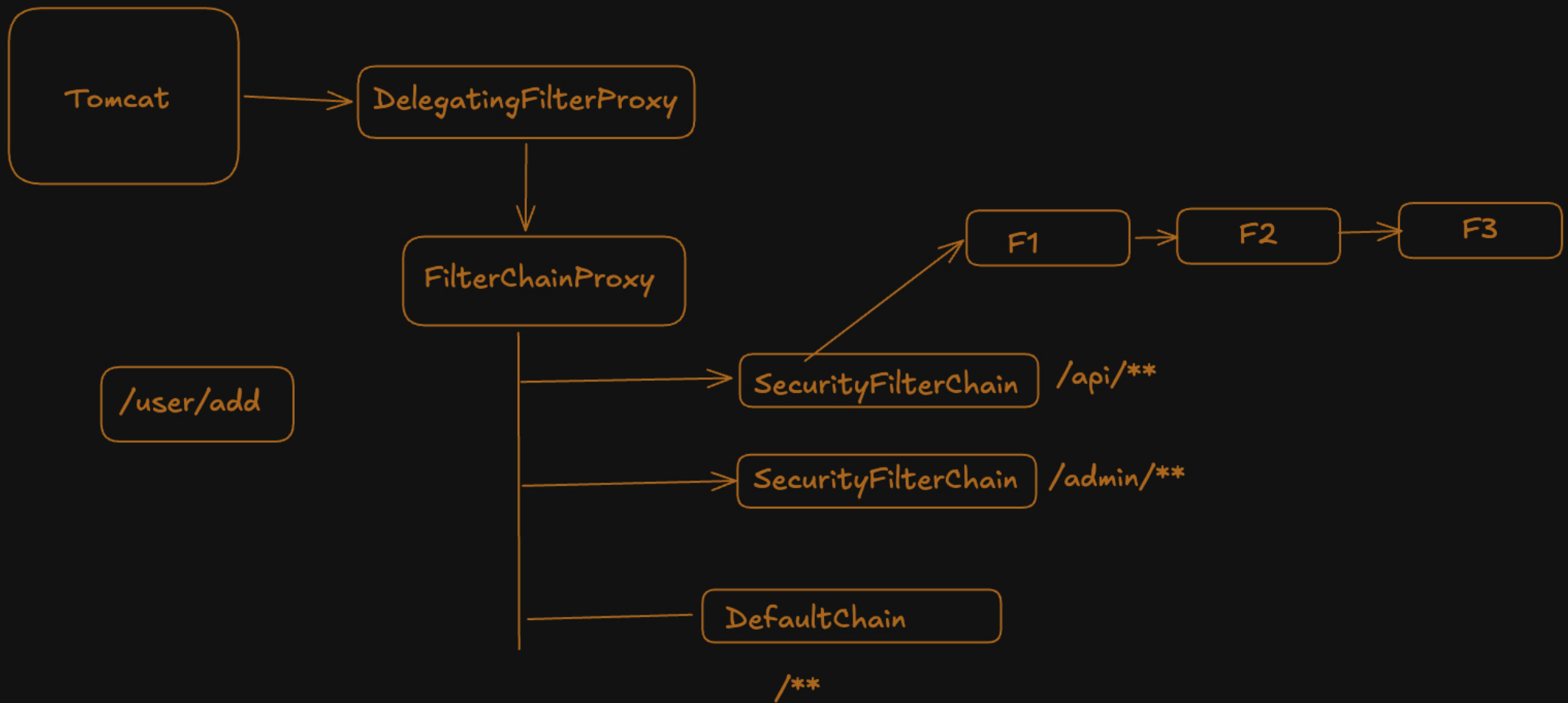
```
public interface UserDetails {  
  
    String getUsername();  
  
    String getPassword();  
  
    Collection<? extends GrantedAuthority> getAuthorities();  
  
    boolean isAccountNonExpired();  
  
    boolean isAccountNonLocked();  
  
    boolean isCredentialsNonExpired();  
  
    boolean isEnabled();  
}
```



1. Take incoming authentication request
2. Ask `UserService` for user
3. Get stored encoded password
4. Ask `PasswordEncoder` whether raw password matches
5. Check account status
6. If successful, return authenticated `Authentication` object







SecurityFilterChain 1

matcher : `/api/**`

Filters : [
 Filter1
 Filter2
 Filter3
]

SecurityFilterChain 2

matcher : `/**`

Filters : [
 Filter1
 Filter2
 Filter3
]


```
public interface SecurityFilterChain {  
    boolean matches(HttpServletRequest request);  
    List<Filter> getFilters();  
}
```

HttpSecurity

returns



DefaultSecurityFilterChain

├─ matcher = any request

├─ filters

│ ├─ SecurityContextHolderFilter

│ ├─ CsrfFilter

│ ├─ LogoutFilter

│ ├─ UsernamePasswordAuthenticationFilter

│ ├─ BasicAuthenticationFilter

│ ├─ AnonymousAuthenticationFilter

│ ├─ ExceptionTranslationFilter

│ └─ AuthorizationFilter

Authentication Filters

Form Login :

→ UsernamePasswordAuthenticationFilter

POST /login

Content-Type : application/x-www-form-urlencoded

/login?username=aditya&password=secret123

HTTP Basic

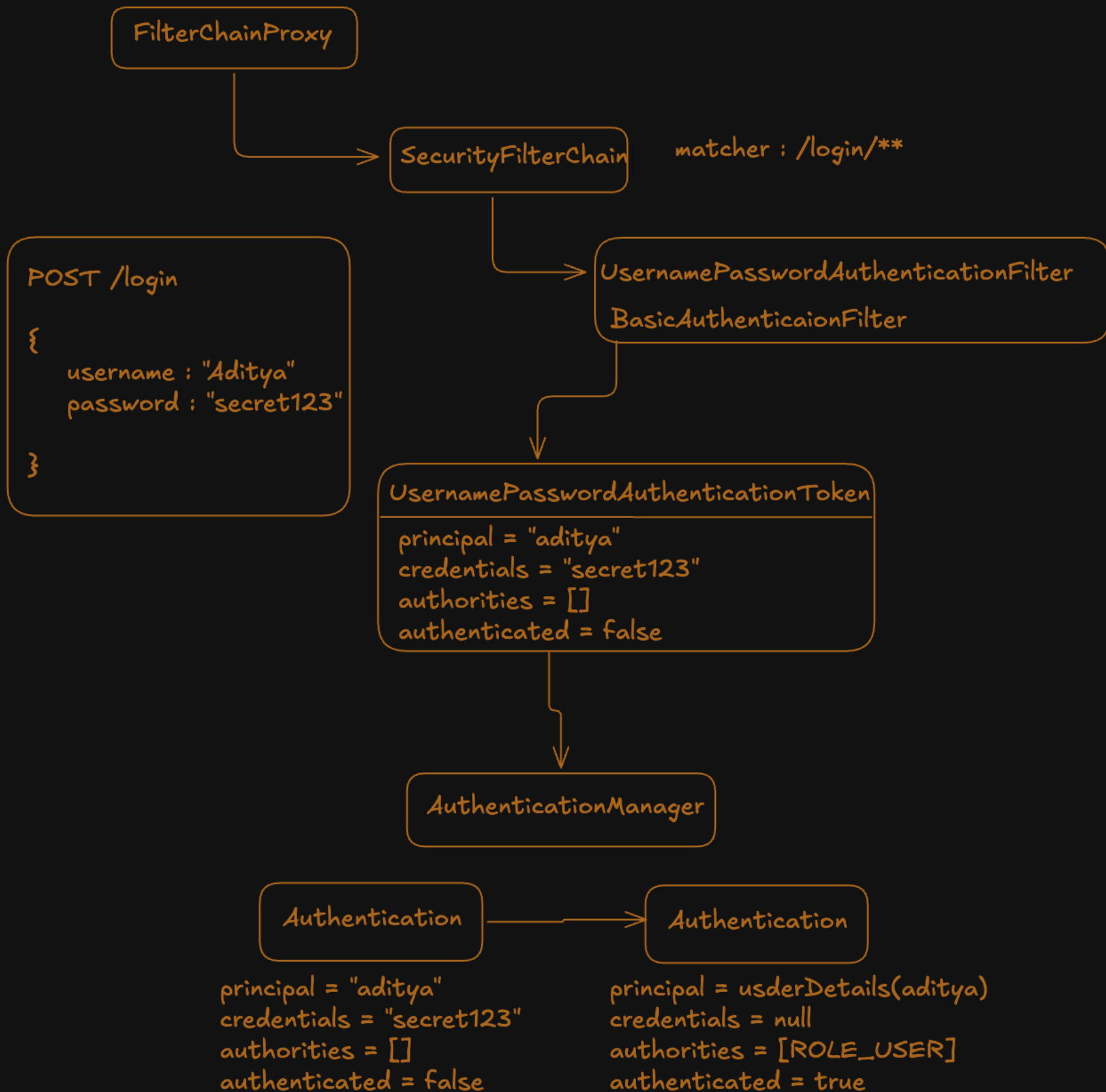
→ BasicAuthenticaitonFilter

POST /login

content-type : application/json

Authorization : Basic Yawersadf==

/login?username=aditya&password=secret123



```
public interface AuthenticationManager {  
    Authentication authenticate(Authentication auth);  
}
```

